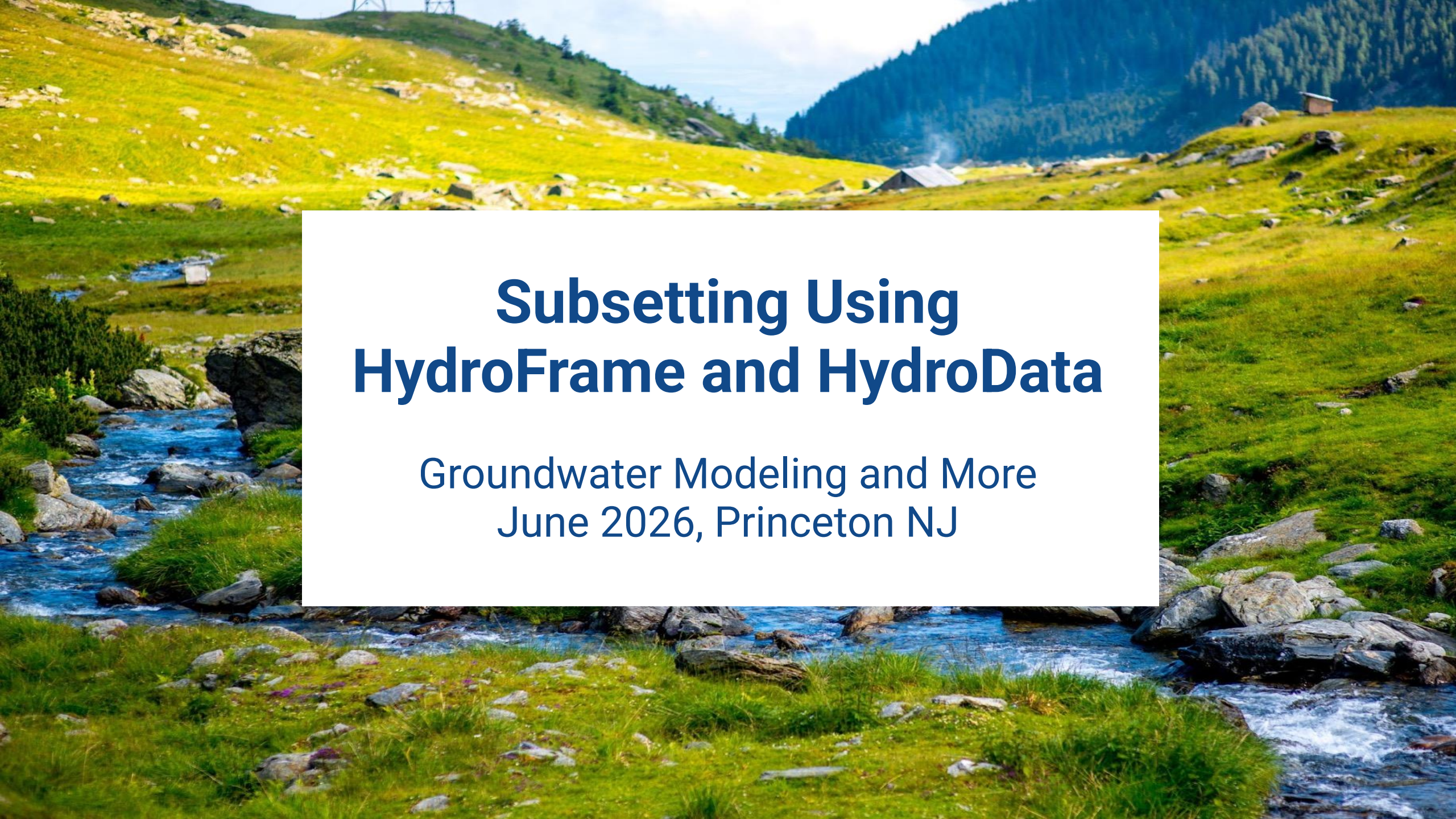




Subsetting Using HydroFrame and HydroData

Groundwater Modeling and More
June 2026, Princeton NJ



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Agenda

1. **Overview of PriorityFlow for Topographic Processing**
2. Overview of HydroFrame and HydroData Toolset
3. SubsetTools Package Overview and Activity
4. HydroData Package Overview and Activity

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1. Overview of PriorityFlow for Topographic Processing
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HydroFrame

- A framework for modeling and data tools related to the national modeling platform
- Our goal is to make national hydrologic simulations and datasets more accessible.

 **HydroFrame** [About](#) [Applications](#) [Data and Simulations](#) [Modeling Tools](#) [Educational Tools](#)



HydroGEN
APPLICATIONS
Explore current and future watershed conditions across the US with HydroGEN



HydroData
DATA AND SIMULATIONS
Access Hydrologic Datasets and Model Results



ParFlow CONUS1.0
DATA AND SIMULATIONS
Learn more about the first generation national ParFlow model



ParFlow CONUS2.0
DATA AND SIMULATIONS
Learn more about the second



ParFlow Resources
MODELING TOOLS
Find training resources and



SubsetTools
MODELING TOOLS
Build your own watershed model

<https://hydroframe.org/>



HydroData: Data Catalog

A database of continuously updated hydrologic simulations and observations that is housed at the Princeton HydroData center

1. **Point observations:** Point observations of hydrologic variables (e.g. streamflow and water table depth) that have been assembled and processed from a variety of external sources.
2. **ParFlow simulation outputs:** Gridded outputs from ParFlow simulations of the contiguous US.
3. **Meteorological forcings:** Gridded datasets that contain meteorological forcing variables (e.g. precipitation, temperature, and wind speed) needed to drive hydrologic models.
4. **Static model inputs and domain files:** Gridded datasets that contain static model inputs that are used directly in model definition (e.g. soil properties, land cover, and hydraulic conductivity) as well as additional domain files that are used in pre and post processing (e.g. distance to streams and watershed boundaries).
5. **Current conditions:** Current hydrologic conditions generated by the [HydroGEN project](#)
6. **Remote sensing:** Gridded observation products that are generated from remote sensing.

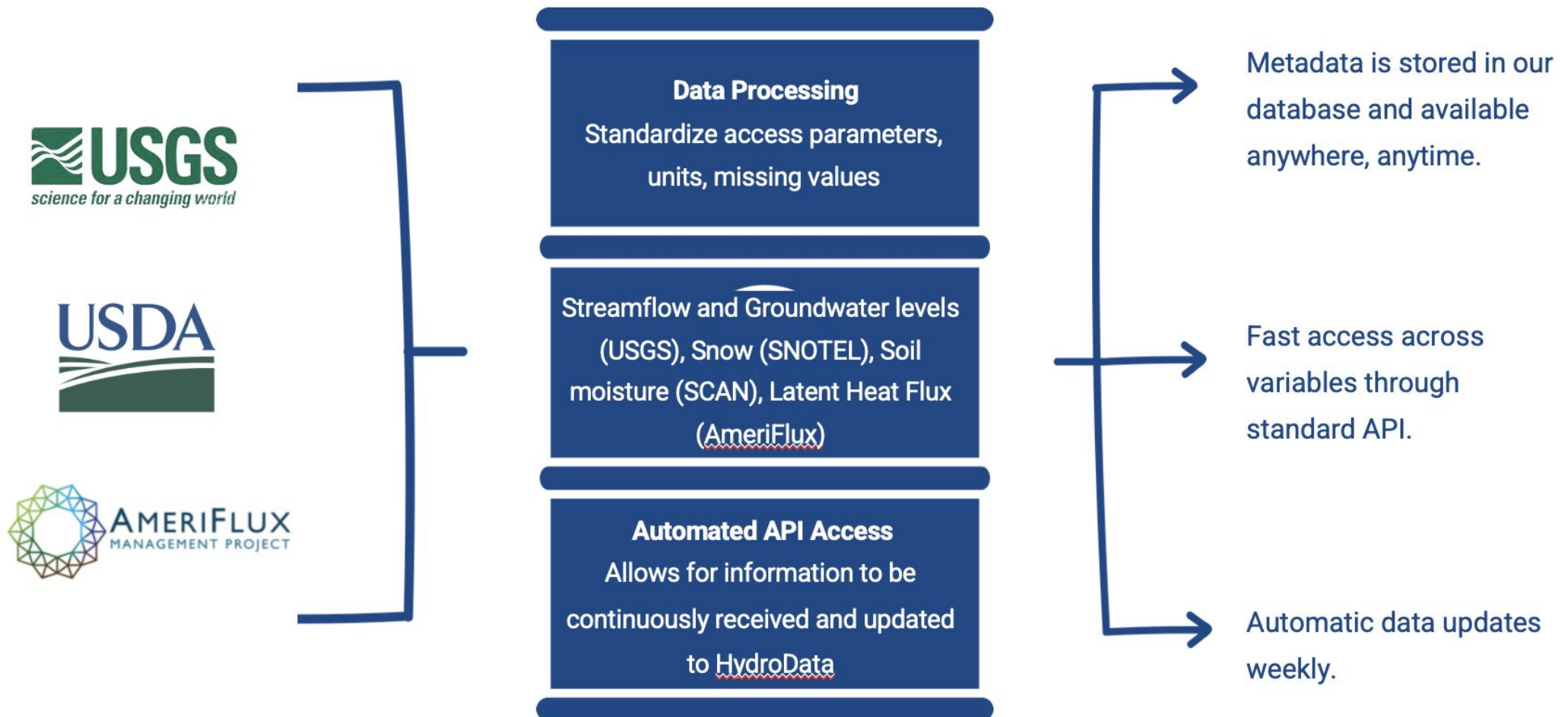


HydroData: Point Observations

Point observations will be the focus of this Short Course



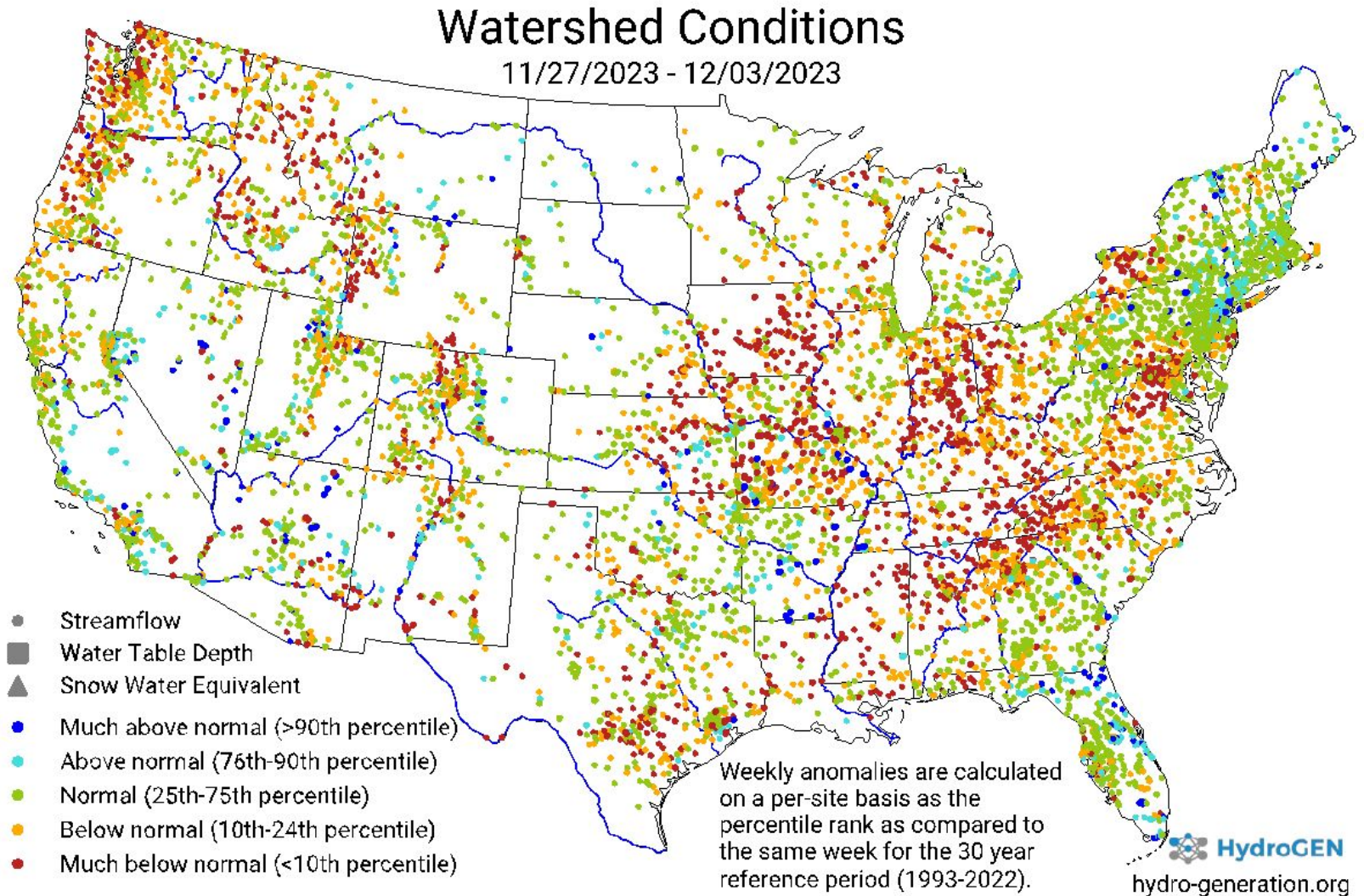
Continuously updated repository of national hydrologic observations





HydroData: Point Observations

By combining data from multiple observation networks we can explore up-to-date watershed conditions across the US considering groundwater, surface water, and snowpack together.

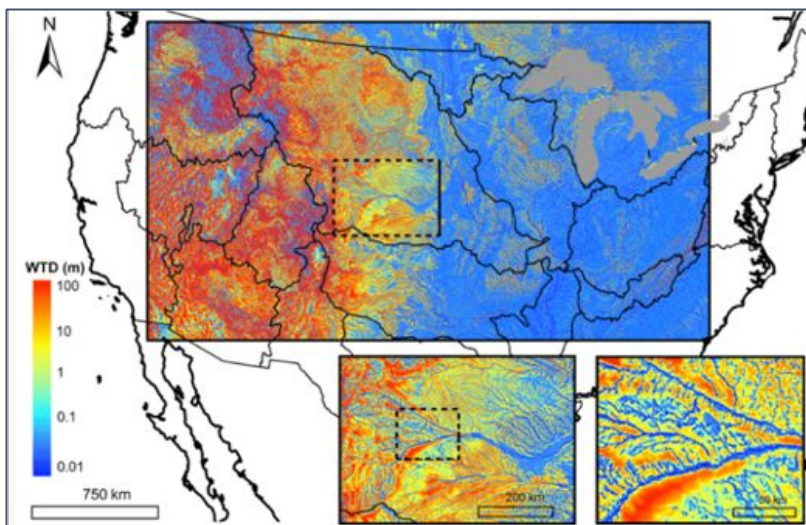




HydroData: Model Datasets

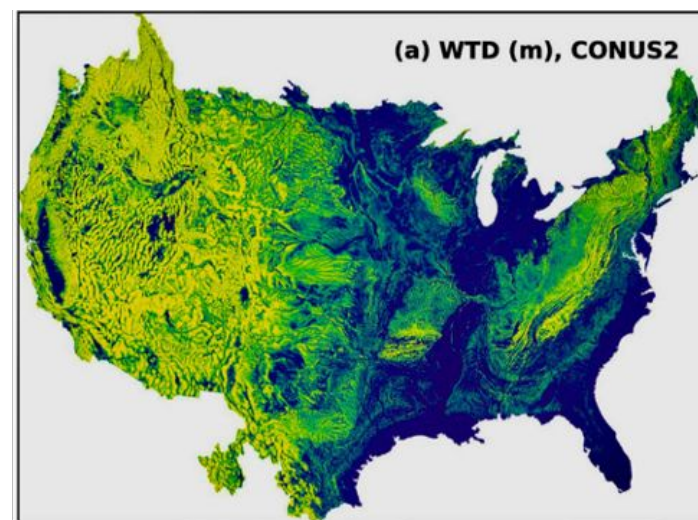
See [our documentation](#) for more information on these datasets

CONUS1 Datasets



- Complete model geofabric
- Steady state water table depth
- Multi-year transient simulation outputs
- NLDAS2 forcing used to run the CONUS1 simulations
- ParFlow and CLM driver files

CONUS2 Datasets



- Updated geofabric for expanded CONUS2 domain
- Steady state water table depth
- CW3E forcing used to run the CONUS2 transient simulations
- ParFlow and CLM driver files



HydroData: Tools

- [hf_hydrodata](#) is a Python package that provides a streamlined syntax for accessing data from **HydroData**
- The [hf_hydrodata](#) package includes functions to access and subset data, return dataset-level or site-level metadata, and relevant dataset citations



hf_hydrodata

The `hf_hydrodata` Python package is a product of the [HydroFrame project](#) and is designed to provide easy access to national hydrologic simulations generated using the National ParFlow model ([ParFlow-CONUS1](#) and [ParFlow-CONUS2](#)) as well as a variety of other gridded model input datasets and point observations. Some of the datasets provided here are direct observations (e.g. USGS streamflow observations) while other are model outputs (e.g. ParFlow-CONUS2) or data products (e.g. remote sensing products).



HydroData: Tools

- Our documentation on [ReadtheDocs](#) includes information on the datasets that are available as well as full examples for how to access and work with the data

The screenshot displays the HydroData documentation interface. On the left is a dark sidebar with the 'hf_hydrodata' logo and a 'latest' version indicator. Below this is a search bar labeled 'Search docs' and a 'TABLE OF CONTENTS' section listing: 'Getting Started', 'Available datasets and data products', 'Working with Gridded Data', 'Working with Point Observations', and 'Python API Reference'. The main content area has a breadcrumb 'Home / HydroData' and a large 'HydroData' heading. The introductory text states: 'The HydroData data catalog, associated python functions `hf_hydrodata`, and API are products of the [HydroFrame project](#) and are designed to provide easy access to national hydrologic simulations generated using the National ParFlow model ([ParFlow-CONUS1](#) and [ParFlow-CONUS2](#)) as well as a variety of other gridded model input datasets and point observations. Some of the datasets provided here are direct observations (e.g. USGS streamflow observations) while other are model outputs (e.g. ParFlow-CONUS2) or data products (e.g. remote sensing products).' To the right of the text is a 'Notebooks' section titled 'Point Observation Notebooks' containing three cards: 'Access and cite point observations data', 'Explore point data availability', and 'Plotting returned point data'. Each card includes a brief description and icons for a video player and GitHub. A pagination indicator shows the first of four items is active.



SubsetTools

- HydroData provides access to CONUS1 and CONUS2 model configurations and outputs including subsetting functionality.
- However, assembling a ParFlow model is very complex and can still pose a barrier to entry for many users
- **Many users want an easy way to accelerate the domain creation process including input data**

```
model.Geom.Perm.Names = "domain s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 g1 g2 g3"
model.Geom.domain.Perm.Type = "Constant"
model.Geom.domain.Perm.Value = 0.02

model.Geom.s1.Perm.Type = "Constant"
model.Geom.s1.Perm.Value = 0.269022595

model.Geom.s2.Perm.Type = "Constant"
model.Geom.s2.Perm.Value = 0.043630356

model.Geom.s3.Perm.Type = "Constant"
model.Geom.s3.Perm.Value = 0.015841225

model.Geom.s4.Perm.Type = "Constant"
model.Geom.s4.Perm.Value = 0.007582087

model.Geom.s5.Perm.Type = "Constant"
model.Geom.s5.Perm.Value = 0.01818816

model.Geom.s6.Perm.Type = "Constant"
model.Geom.s6.Perm.Value = 0.005009435

model.Geom.s7.Perm.Type = "Constant"
model.Geom.s7.Perm.Value = 0.005492736

model.Geom.s8.Perm.Type = "Constant"
model.Geom.s8.Perm.Value = 0.004675077

model.Geom.s9.Perm.Type = "Constant"
model.Geom.s9.Perm.Value = 0.003386794

model.Geom.s10.Perm.Type = "Constant"
model.Geom.s10.Perm.Value = 0.004783973
```



SubsetTools

Help users build watershed models from the national ParFlow modeling framework

- Functions that simplify the process of setting up a modeling domain and running ParFlow
- Allows the user to subset all required hydrogeologic and climate forcing datasets from the HydroData filesystem
- Template runscripts for common use cases and examples for how to use them
- Helper functions to customize and run your subset model



SubsetTools

Workbooks and tutorials provide end-to-end workflows to build and run steady state and transient ParFlow models

subsettools
latest

Search docs

Getting started

How To

Example notebooks

API Reference

Changelog

Contributing

Code of Conduct

Data Use Policy

SubsetTools

Introduction

The **SubsetTools** package is developed and maintained by the **HydroFrame** project. It is designed to subset model inputs and outputs from the national ParFlow modeling framework. We provide tools

Conus1 SubsetTools

Define a subset domain area

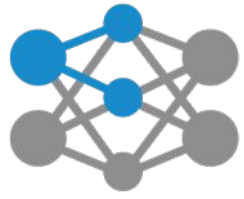
Illustrates three different ways to define a subset domain.

Subset Gridded Static Inputs

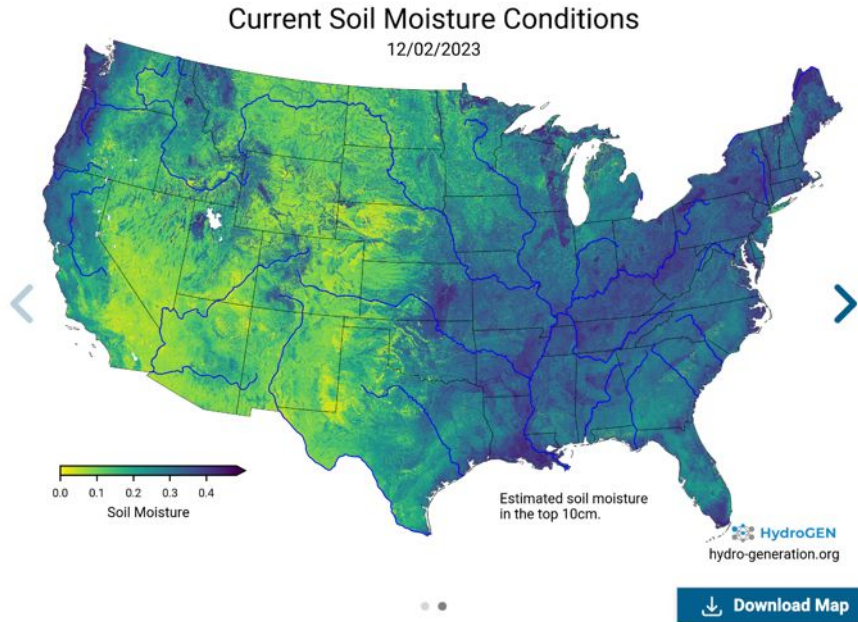
Two different approaches for subsetting static input variables and working with the data

Subset Gridded Forcing Data

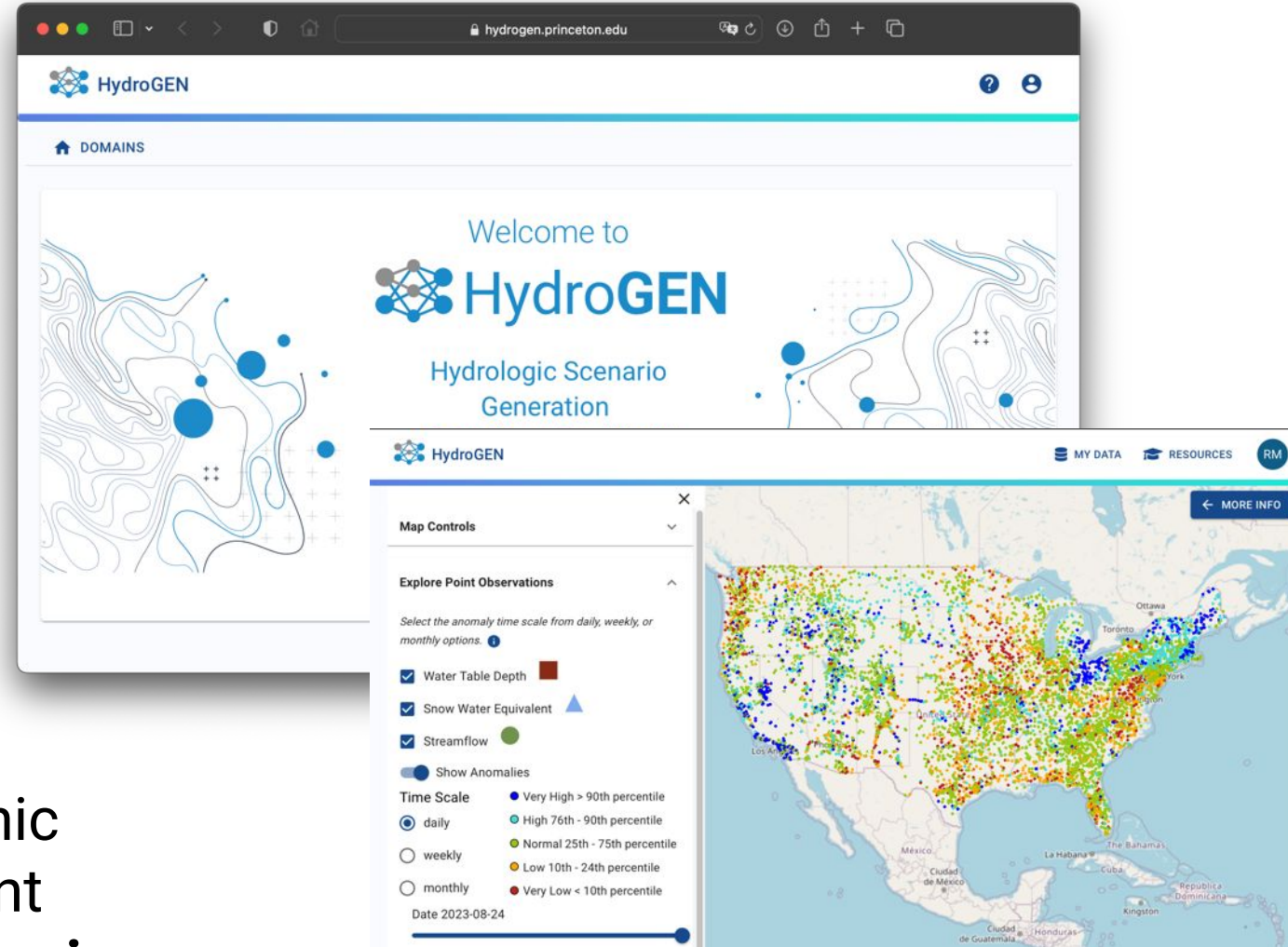
Two different approaches for subsetting meteorological forcing variables and working with the data



HydroGEN



A user friendly app to explore geographic domains, point observations and current conditions. **No modeling or Python experience required.**



hydro-generation.org

Agenda

1. Overview of PriorityFlow for Topographic Processing
2. Overview of HydroFrame and HydroData Toolset
3. **SubsetTools Package Overview and Activity**
4. HydroData Package Overview and Activity

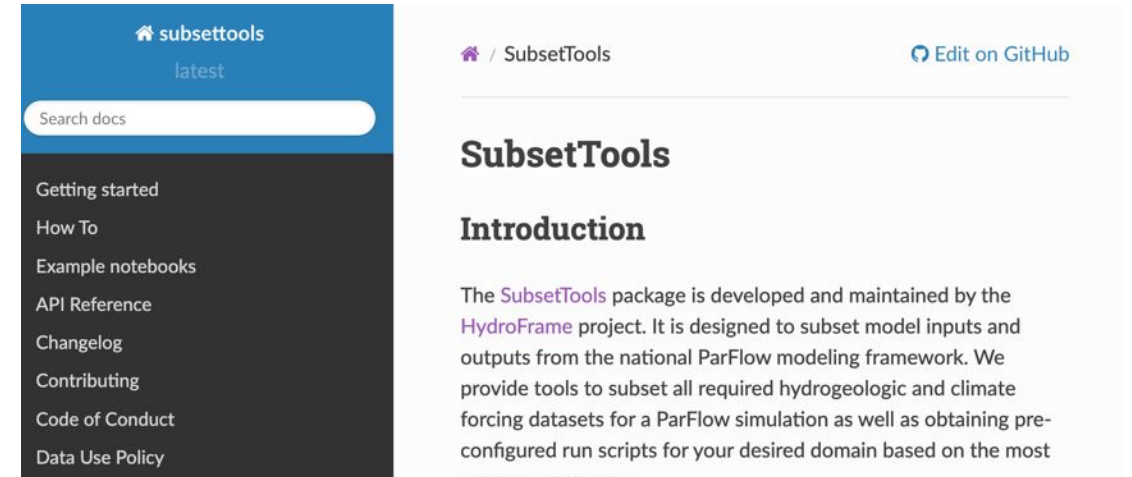


SubsetTools: Usage and Installation

Getting access to the HydroData Catalog and SubsetTools is easy!

1. [Sign up](#) for a HydroFrame account
2. [Get a PIN](#) for API access
3. Install the SubsetTools package

```
$ pip install subsettools
```



Everything is documented on our [Read the Docs page](#)
(you can find the link to this on hydroframe.org)

Ways to Interact with SubsetTools

Resources

- GitHub CodeSpaces
 - Launch directly from the ShortCourse GitHub repository
- [Binder](#)
 - Easiest: ParFlow and subsettools already installed, tutorials included and runnable
 - Limited resources (max 6 hours session, 1 CPU hour total)
 - Basic ParFlow version (no support for parallel runs, no optional packages)
- Docker:
 - Container with subsettools and ParFlow installed.
 - Linux version supported for both arm64 and amd64 architectures
 - Installing Docker is much easier than installing ParFlow
- Pip install in a virtual environment
 - Needs some knowledge about virtual environments
 - If a user wants to do a simulation as well, needs ParFlow installed and some environment variables configured



SubsetTools: Tutorials

We have detailed tutorial walking through examples of common workflow needs for

- Define a domain of interest using a bounding box or a watershed boundary
- Accessing model input files to configure a domain
- Working with atmospheric forcing data
- Accessing ParFlow and CLM template run scripts and configuring them for a custom domain/simulation

☐ How To

Define a subset domain area

Creating solid files

Subset Gridded Static Inputs

Subset Gridded Forcing Data

Configure CLM inputs from template

Build a ParFlow run from a template



SubsetTools: Example Workflows

We currently have three end-to-end workflows for subsetting and running ParFlow simulations including steady state and transient examples for CONUS1 and CONUS2

1. **Performing a model initialization (spin up) with ParFlow on CONUS1:** This notebook will subset a HUC8 from the CONUS1 domain and will create a ParFlow run script to run a spinup simulation uses a constant recharge forcing to achieve a steady state water table depth (NOTE: This run does not include land surface simulation).
2. **Transient simulation with ParFlow-CLM on CONUS1:** This notebook walks through an example of subsetting a HUC8 from the CONUS1 domain. This example will subset everything needed to do a transient run with ParFlow-CLM and create a run script. This simulation is designed to start from a steady state groundwater condition. Here we subset the initial conditions from the pre-generated national steady state water table depth simulations, but the starting point could also be generated from the spin up workbook.
3. **Transient simulation with ParFlow-CLM on CONUS2:** This notebook walks through an example of subsetting a HUC8 from the CONUS2 domain. This example will subset everything needed to do a transient run with ParFlow-CLM and create a run script. This simulation is designed to start from a steady state groundwater condition. Here we subset the initial conditions from the pre-generated national steady state water table depth simulations, but the starting point could also be generated from the spin up workbook.

Hands-on activity 1

We will run a couple of examples to get familiar with common workflow steps:

- Defining a subset domain
- Getting forcing data from HydroData
- Launch CodeSpaces
- subsettools/definte_subset.ipynb
- subsettools/subset_forcing_data.ipynb
- First try running the notebooks as they are. Then, try modifying them to define your own subset domain and/or start and end dates for your forcing data.

Walk through of a subsetting workflow

Subset CONUS and run ParFlow-CLM

To launch this notebook interactively in a Jupyter notebook-like browser interface, please click the “Launch Binder” button below. Note that Binder may take several minutes to launch.



This notebook walks through an example of subsetting a HUC8 from the CONUS2 domain. This example will subset everything needed to do a transient run with ParFlow-CLM. This includes all hydrogeologic datasets as well as climate forcing data from CW3E. All of the data is written to a folder for the specified days to run. This example uses the template runscript `conus2_pfclm_transient_solid.yaml` and edits it to correspond with the domain subset. It also sets up and performs the designed simulation. The result will be model output pressure and saturation `pfbs` according to the days specified.

This notebook has two principal sections:

1. Subset all static inputs and climate forcings from a CONUS run stored in Hydrodata
2. Load and alter a reference run to set up and perform your ParFlow-CLM subset.

Import the required libraries

```
import matplotlib.pyplot as plt
import numpy as np
import os
from parflow import Run
```

Hands-on activity 2

Now you will try running the subsetter workflow.

- Launch CodeSpaces
- `subsettools/conus2_subsetting_transient.ipynb`
- First you can run it as is then modify to select a different HUC and/or start and end dates.

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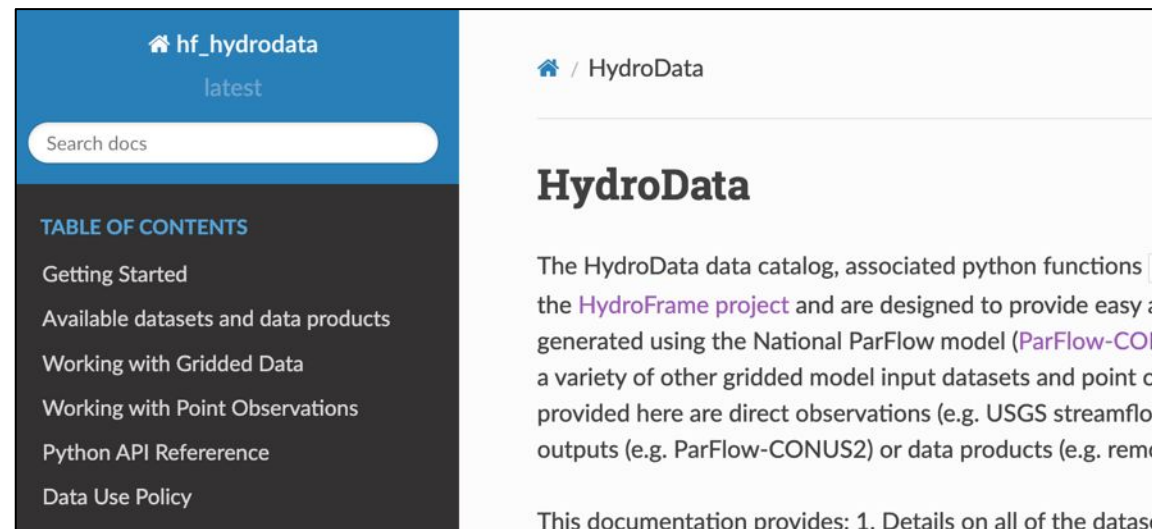


hf_hydrodata: Usage and Installation

Similar to the Subsettools package:

1. [Sign up](#) for a HydroFrame account
2. [Get a PIN](#) for API access
3. Install the hf_hydrodata package

```
$ pip install hf_hydrodata
```



Everything is documented on our [ReadtheDocs](#) page



HydroData: Exploring the Catalog

All available datasets as well as variable definitions, grid coordinates, and metadata definitions are defined on the HydroData [ReadtheDocs](#)

TABLE OF CONTENTS

Getting Started

☐ Available datasets and data products

Datasets

Variables

Grids

Temporal Resolutions

Aggregations

Metadata

Available datasets and data products

The HydroData catalog contains a variety of datasets and data products to support national hydrologic modeling and analysis. We have the following categories of datasets:

1. **Point observations:** Point observations of hydrologic variables (e.g. streamflow and water table depth) that have been assembled and processed from a variety of external sources.
2. **ParFlow simulation outputs:** Gridded outputs from ParFlow simulations of the contiguous US.
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5. **Current conditions:** Current hydrologic conditions generated by the [HydroGEN project](#)
6. **Remote sensing:** Gridded observation products that are generated from remote sensing.



HydroData: Exploring the Catalog

Every dataset has its own documentation page providing a detailed description of how it was generated and all available variables

Modern CONUS1 simulations WY 2003-2006

A simulation of water years 2003-2006 completed with the ParFlow CONUS1 model. This is the most recent transient simulation that was completed with the ParFlow CONUS1 model (i.e. after the [conus_baseline_85](#) simulation). The simulation was completed using the NLDAS2 forcing dataset at 1km resolution and a 1 hour time step. This dataset contains all of the outputs from the simulation as well as the ParFlow and CLM run scripts. The model inputs can be found in the [conus1_domain](#) dataset. Refer to the citations below for additional information on the model development and findings.

Dataset Name: conus1_baseline_mod

Data Source: hydroframe

Citations:

Please refer to the following citations for more information on this dataset and cite them if you use the data

- <https://doi.org/10.5194/gmd-14-7223-2021>

Extent and Resolution:

- Available Date Range: 10/1/02 to 9/30/06
- Grid: conus1
 - Spatial Resolution: 1000 meters

variable	description	temporal_resolut
evapotranspiration	Evapotranspiration	daily, hourly
latent_heat	Latent heat flux from canopy height to atmosphere	hourly
outward_longwave_radiation	Outgoing long-wave radiation from ground+canopy	hourly
sensible_heat	Sensible heat from canopy height to atmosphere	hourly
ground_heat	Ground heat flux	hourly
ground_evap	Ground surface evaporation rate	hourly
ground_evap_heat	Evaporation heat flux from ground	hourly
transpiration_leaves	Evaporation+transpiration from leaves	hourly
transpiration	Transpiration rate	hourly

Run File Variables

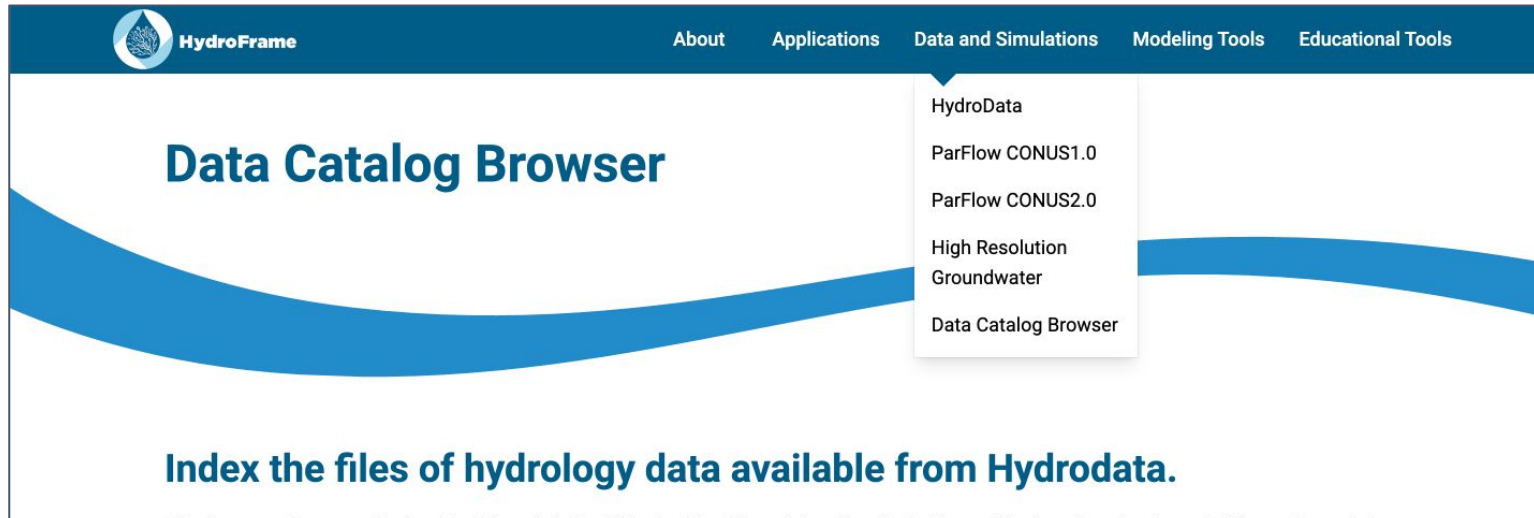
variable	description	temporal_resolution	units
parflow_run	Run scripts and other model configuration files	static	unitless
clm_run	No description	static	unitless

Subsurface Variables

variable	description	temporal_resolution	units
soil_moisture	Soil moisture	daily	kg/m2
water_table_depth	Water table depth	daily	m
pressure_head	Map 3d grid point to pressure head	daily, hourly	m
pressure_next	Pressure head at the following timestep	daily	m
saturation	Map 3d grid point to saturation value	hourly	unitless



HydroData: Exploring the Catalog



Data Catalog Entries			
Dataset	Variable	Units	Grid
CW3E	air_temp	K	conus2
CW3E	atmospheric_pressure	pa	conus2
CW3E	downward_longwave	w/m2	conus2
CW3E	downward_shortwave	w/m2	conus2
CW3E	east_windspeed	m/s	conus2
CW3E	north_windspeed	m/s	conus2

<https://hydroframe.org/datacatalog-browser>



HydroData: Point Data Tools

We have a series of example notebooks illustrating how to subset, analyze and cite point observations across the HydroData Catalog

Filter sites by pre-defined site networks

Walk through how to filter point observations data to only sites within pre-defined networks such as GAGES-II or CAMELS.



Explore point data availability

Walk through how to explore what data is available.



Access and cite point observations data

Walk through some example functionality for accessing and citing point observations.



Save and wrangle point observations data

Explore some basic features of pandas DataFrames.



Plotting returned point data

Walk through some examples for how to plot data returned from `get_point_data`.



Walk through of an hf_hydrodata workflow

Generate + Code + Markdown | Run All Restart ... Python (Subsetting ShortCourse) (Python 3.12.3)

Explore and request point observations data

This notebook provides a walk-through of some example functionality for exploring and requesting point observations data and site-level metadata from the [HydroData Data Catalog](#) using the `hf_hydrodata` Python package. We will use `hf_hydrodata`'s `get_site_variables`, `get_point_metadata` and `get_point_data` functions. We will also review how to properly attribute/cite the data acquired.

The `hf_hydrodata` documentation has more complete information on which [point observation datasets](#) are supported and what [site metadata features](#) are available. There is also [full API documentation](#) available for all of the `hf_hydrodata` functions. The API documentation is the most up-to-date resource for supported parameters and function options.

Set-Up

```
import pandas as pd
import hf_hydrodata as hf
from matplotlib import pyplot as plt
import plotly.express as px
import plotly.io as pio
```

[1] ✓ 17.9s Python

Hands-on activity 1

Now you will try running a point observations workflow.

- Launch CodeSpaces
- hf_hydrodata/point/**example_get_observations.ipynb**
- First you can run it as-is. Then explore!
 - Try a different HUC
 - Try a different date range
 - Try to get variable=water_table_depth from USGS NWIS

Hands-on activity 2

Option 1

Want to explore how to visualize outputs from the point observations module?

- hf_hydrodata/point/**example_plot_data.ipynb**

Option 2

Want to get more familiar with how to work with pandas DataFrames?

- hf_hydrodata/point/**example_pandas.ipynb**



THANK YOU